

Full Stack Development and Deployment of a Web-Based Speech Recognition System

Weeks 31-32
10 August - 23 August 2026

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Project Overview

- MAGOR is a web-based audio and video transcription system that converts recordings into text
- **Existing Problem:**
 - Although functional, components remain incomplete and require refinement
- **Proposed Solution:**
 - Improve robustness, usability, and AI-driven recording analysis



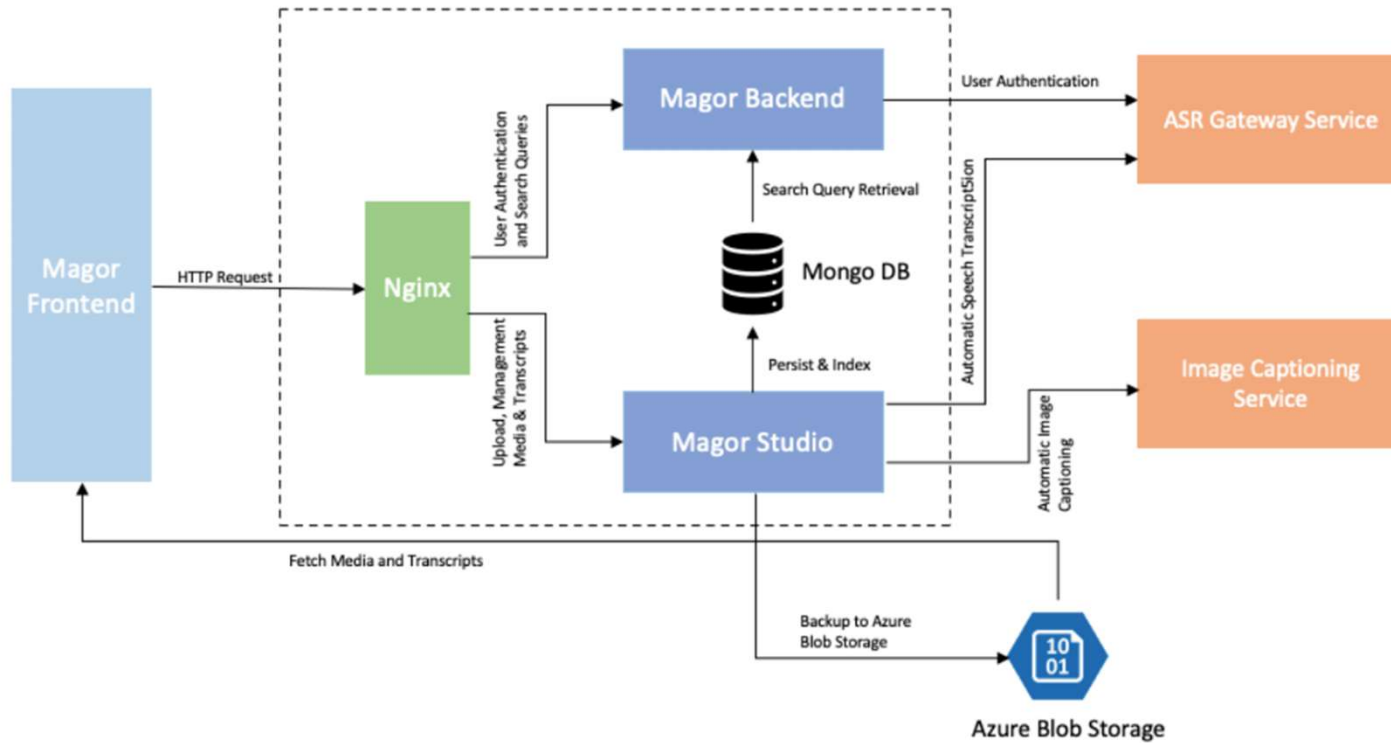
Project Objectives

The primary objectives of this project are:

- Fix system bugs and restore functionality
- Improve usability and user experience
- Enhance maintainability and scalability
- Add AI features to existing pages



Systems Overview



Source: Huang (2021)



Agenda

Work Done Over The Weeks:

- “Ask AI” Button functionality
 - Implemented Q&A tab question-and-answer functionality
 - Added persisted storage and retrieval of previous Q&A for each recording
 - Manual polling after automatic polling times out or user navigates away
 - Contextual Q&A: Previous Q&A exchanges used as context when answering new questions



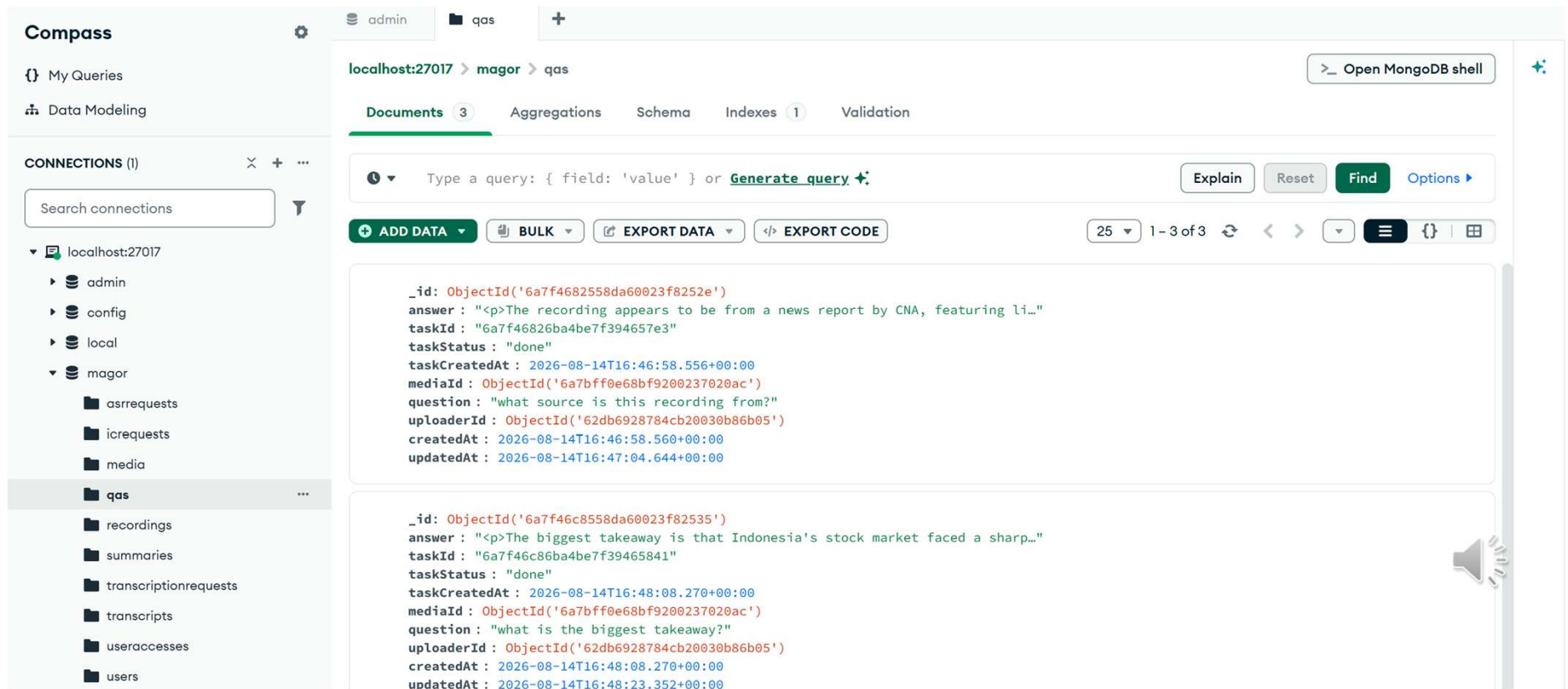
Task 1 (Q&A Functionality)

- No Q&A UI:

The screenshot displays the Magor application interface. At the top, a blue header bar contains the 'Magor' logo, a search bar, a dropdown menu set to 'This recording', and a 'test' button. Below the header, the main content area is titled 'Original_recording with summary'. On the left, a video player shows a man in a dark suit speaking at a podium. The video player controls at the bottom indicate '0 matches' and 'Nothing is playing right now.' with a progress bar at 0:00 / 0:13. On the right, a sidebar contains a 'Transcript' tab, an 'Ask AI' button, and an 'Edit' button. Below these, there are tabs for 'Summarize' and 'Q&A'. The 'Q&A' tab is active, showing a text input field with the placeholder 'Ask a question about the transcript...' and an 'Ask' button. At the bottom of the sidebar, a message states 'No questions yet' and 'Ask something about the transcript to see an answer here.' A speaker icon is visible in the bottom right corner of the interface.

Task 1 (Q&A Functionality)

- MongoDB qa implementation:



The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (1)' panel lists the connection 'localhost:27017' with a tree view showing databases 'admin', 'config', 'local', and 'magor'. Under 'magor', several collections are listed, including 'qas' which is currently selected. The main panel displays the 'qas' collection with 3 documents. The 'Documents' tab is active, showing two documents in a list view. Each document contains fields for _id, answer, taskId, taskStatus, taskCreatedAt, mediaId, question, uploaderId, createdAt, and updatedAt. The first document's answer is 'The recording appears to be from a news report by CNA, featuring li...' and the second document's answer is 'The biggest takeaway is that Indonesia's stock market faced a sharp...'. A 'Generate query' button is visible in the query bar, and a speaker icon is in the bottom right corner of the interface.

Compass

admin qas +

localhost:27017 > magor > qas

Documents 3 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain Reset Find Options

ADD DATA BULK EXPORT DATA EXPORT CODE

25 1 - 3 of 3

```
_id: ObjectId('6a7f4682558da60023f8252e')
answer: "<p>The recording appears to be from a news report by CNA, featuring li..."
taskId: "6a7f46826ba4be7f394657e3"
taskStatus: "done"
taskCreatedAt: 2026-08-14T16:46:58.556+00:00
mediaId: ObjectId('6a7bff0e68bf9200237020ac')
question: "what source is this recording from?"
uploaderId: ObjectId('62db6928784cb20030b86b05')
createdAt: 2026-08-14T16:46:58.560+00:00
updatedAt: 2026-08-14T16:47:04.644+00:00
```

```
_id: ObjectId('6a7f46c8558da60023f82535')
answer: "<p>The biggest takeaway is that Indonesia's stock market faced a sharp..."
taskId: "6a7f46c86ba4be7f39465841"
taskStatus: "done"
taskCreatedAt: 2026-08-14T16:48:08.270+00:00
mediaId: ObjectId('6a7bff0e68bf9200237020ac')
question: "what is the biggest takeaway?"
uploaderId: ObjectId('62db6928784cb20030b86b05')
createdAt: 2026-08-14T16:48:08.270+00:00
updatedAt: 2026-08-14T16:48:23.352+00:00
```

Task 1 (Q&A Functionality)

- Manual Polling:

Magor

Search

This recording

test

Edit

Indonesian stocks plunge more than 7 after MSCI flags risks



0 matches

Nothing is playing right now.

Transcript

Ask AI

Summarize

Q&A

Ask a question about the transcript...

Ask

Still answering: "How relevant is this news?"

Check Status

Previous Q&A

Q: What could be the potential consequences of Indonesia being downgraded to frontier market status?

A: A downgrade to frontier market status could lead to a flight of capital from Indonesia, as global investors may reassess the market's accessibility and perceive it as riskier compared to emerging markets (01:36–01:44, 02:57–03:00). This could result in reduced foreign investment, increased market volatility, and further erosion of investor confidence due to concerns over transparency, fiscal discipline, and central bank independence (02:24–02:27).

Q: What changes does Indonesia need to make to avoid being downgraded by MSCI?

A: To avoid being downgraded by MSCI, Indonesia needs to make sufficient progress on improving market investability, particularly by addressing concerns over thin trading, concentrated ownership in major companies, and lack of transparency in corporate ownership structures (00:15–00:22). The Indonesia Stock Exchange (IDX) must also resolve issues related to the free float calculation of domestic shares, as the changes made so far have been deemed

Task 1 (Q&A Functionality)

- Past Q&A Context:

```
8 // build a context string from past Q&A pairs
9 // starting from the most recent and going backwards
10 // until we reach the character limit
11 function buildQAContext(pastQA: { question: string; answer: string }[]): string {
12   const lines: string[] = [];
13   let total = 0;
14   for (let i = pastQA.length - 1; i >= 0; i--) {
15     const qa = pastQA[i];
16     const entry = `Q: ${qa.question}\nA: ${qa.answer}`;
17     if (total + entry.length > MAX_QA_CONTEXT_CHARS) break;
18     lines.unshift(entry);
19     total += entry.length;
20   }
21   return lines.join('\n\n');
22 }
```

Build Q&A Context Function

```
const requestPayload = {
  taskType: "SUM",
  dialogue: transcript,

  userPrompt: `You are answering questions about the transcript above,
in the style of YouTube's Ask AI.
```

Context

```
`${context ? `Here is the earlier conversation on this recording for context:
${context}
` : ''}
```

Answer this question using only information from the transcript
(and the conversation above, if relevant):

```
"${question}"
```

Question

Keep the answer to a few sentences, and include supporting timestamps
in parentheses where relevant, for example (01:32-03:45).

```
If the transcript doesn't contain the answer, say so plainly -
never invent information.`;
};
```

Payload: Context and Question in userPrompt

Task 1 (Q&A Functionality)

- Q&A Functionality Demo:

The screenshot displays the Magor application interface. At the top, a blue header bar contains the 'Magor' logo, a search bar, and a 'This recording' dropdown menu. Below the header, the main content area features a video player on the left and a Q&A sidebar on the right.

Video Player: The video player shows a news broadcast titled 'ASIA Business Tonight' with a background image of the Petronas Towers. Below the video, a search bar indicates '0 matches' and a message states 'Nothing is playing right now.' The video progress bar shows '0:00 / 3:50'.

Q&A Sidebar: The sidebar is titled 'Transcript' and 'Ask AI'. It contains a 'Summarize' button and a 'Q&A' button. The 'Q&A' section displays a summary of the video content, including a paragraph about Indonesian shares and a list of key points.

Summary: Indonesian shares plunged nearly 9%, triggering a circuit breaker, amid concerns over market investability raised by MSCI, which warned of a potential downgrade to frontier market status by May 2026 if transparency and trading issues are not addressed. Investor confidence is further shaken by questions about central bank independence and fiscal discipline following recent political appointments and policy moves (00:00-03:39).

Key points:

- **Market plunge and circuit breaker:** Indonesian shares dropped nearly 9% at one point, triggering a 30-minute trading halt, with the Jakarta composite closing down over 7% (00:00-00:11).
- **MSCI investability warning:** MSCI flagged concerns over thin trading and ownership concentration in major Indonesian companies, threatening a downgrade to frontier market status if reforms aren't made by May 2026 (00:11-00:26).
- **Fiscal and central bank concerns:** Investor confidence is undermined by worries over fiscal discipline and Bank Indonesia's independence, especially after surprise rate cuts.

At the bottom of the sidebar, there is a 'Generate New Summary' button.

Plans for Next Update

- Fix timestamp navigation issues
 - Resolve non-working timestamp jumps
- Implement Q&A pagination
 - Load recent Q&As first
 - Load older entries on scroll

